



API Specification

Backend REST API Specification & CRUD Documentation

Architecture: RESTful API / Node.js (Express) & PostgreSQL (Prisma)

1. Global API Overview

1.1 Base URLs

Environment	Base URL
Local Development	<code>http://localhost:5000/api/v1</code>
Staging	<code>https://checkpoint-dev.onrender.com/api/v1</code>
Production	<code>https://checkpoint-prod.onrender.com/api/v1</code>

1.2 Authentication & Security

All protected endpoints require a JSON Web Token (JWT) supplied in the standard HTTP `Authorization` header:

```
Authorization: Bearer <JWT_TOKEN>
```

User Roles & Access Control

- ADMIN**: Global system administrator with unrestricted access.
- ORGANIZER**: Event owner with permissions to create events, manage checkpoints, register attendees, view analytics, and export reports.

- **STAFF** : Assigned event crew permitted to operate QR scanner stations and view checkpoint check-in logs.
- **PUBLIC** : Unauthenticated access (e.g., event browsing, public RSVP registration, ticket verification).

1.3 Standard Request Headers

```
Content-Type: application/json
Accept: application/json
```

1.4 Unified Response Schemas

Standard Success Response (200 OK , 201 Created)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Operation completed successfully",
  "data": {}
}
```

Standard Error Response (400 , 401 , 403 , 404 , 409 , 500)

```
{
  "success": false,
  "statusCode": 400,
  "error": "BAD_REQUEST",
  "message": "Specific explanation of the failure condition",
  "details": []
}
```

1.5 HTTP Status Codes Standard

Status Code	Meaning	Usage Scenario
200 OK	Request Succeeded	Successful GET , PUT , PATCH , or DELETE .
201 Created	Resource Created	Successful POST creating an event, registration, or checkpoint.
400 Bad Request	Validation Error	Missing required fields, invalid format, or malformed body.
401 Unauthorized	Authentication Missing	Missing, expired, or invalid JWT token.
403 Forbidden	Permission Denied	Authenticated user lacks required role for the resource.
404 Not Found	Resource Missing	Event, checkpoint, user, or registration ticket code not found.
409 Conflict	State Conflict	Duplicate email registration or single-scan check-in violation.
500 Internal Error	Server Error	Unhandled backend exception or database failure.

2. Authentication & User Management (**/auth** , **/users**)

2.1 User Login

Authenticates an Admin, Organizer, or Staff member and returns a signed JWT.

- **URL:** **/auth/login**
- **Method:** **POST**
- **Access Control:** Public

Request Body

```
{
  "email": "organizer@example.com",
  "password": "SecurePassword123!"
}
```

Success Response (200 OK)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Authentication successful",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "expiresIn": 86400,
    "user": {
      "id": "a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11",
      "email": "organizer@example.com",
      "fullName": "John Doe",
      "role": "ORGANIZER",
      "isActive": true
    }
  }
}
```

Error Response (401 Unauthorized)

```
{
  "success": false,
  "statusCode": 401,
  "error": "UNAUTHORIZED",
}
```

```
{
  "message": "Invalid email or password"
}
```

2.2 Get Current Authenticated Profile

Retrieves account information and event assignments for the active token.

- **URL:** `/auth/me`
- **Method:** `GET`
- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER` , `STAFF`)

Success Response (`200 OK`)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Profile fetched successfully",
  "data": {
    "id": "a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11",
    "email": "organizer@example.com",
    "fullName": "John Doe",
    "role": "ORGANIZER",
    "isActive": true,
    "createdAt": "2026-09-01T08:00:00.000Z"
  }
}
```

2.3 Update Profile

Updates user contact details and display preferences.

- **URL:** `/auth/profile`
- **Method:** `PUT`
- **Access Control:** Authenticated

Request Body

```
{
  "fullName": "John Doe"
}
```

Success Response (200 OK)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Profile updated successfully",
  "data": {
    "id": "a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11",
    "fullName": "John Doe",
    "updatedAt": "2026-10-06T13:00:00.000Z"
  }
}
```

3. Event Management (/events)

3.1 Create Event

Creates a new event with scheduling, location, and configuration settings.

- **URL:** /events
- **Method:** POST
- **Access Control:** Authenticated (ADMIN , ORGANIZER)

Request Body

```
{
  "title": "BSIT Tech Summit 2026",
}
```

```
"slug": "bsit-tech-summit-2026",
"description": "Annual student technology conference and project showcase.",
"venueName": "PGymnasium, Main Campus",
"startDate": "2026-11-15T08:00:00.000Z",
"endDate": "2026-11-15T17:00:00.000Z",
"metadata": {
  "capacity": 500,
  "eventFormat": "MULTI_STOP",
  "allowWalkIns": true
}
```

Success Response (201 Created)

```
{
  "success": true,
  "statusCode": 201,
  "message": "Event created successfully",
  "data": {
    "id": "e2a1496a-04b3-4f98-904d-17bb27a7c501",
    "organizerId": "a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11",
    "title": "BSIT Tech Summit 2026",
    "slug": "bsit-tech-summit-2026",
    "description": "Annual student technology conference and project showcase.",
    "venueName": "Gymnasium, Main Campus",
    "startDate": "2026-11-15T08:00:00.000Z",
    "endDate": "2026-11-15T17:00:00.000Z",
    "isActive": true,
    "createdAt": "2026-10-06T13:50:00.000Z"
  }
}
```

3.2 List Events

Fetches all events with filtering, search, and pagination.

- **URL:** `/events`
- **Method:** `GET`
- **Access Control:** Public (Public events) / Authenticated (Organizer own events)
- **Query Parameters:**
 - `page` (integer, default: `1`): Current page number.
 - `limit` (integer, default: `10`): Items per page.
 - `search` (string, optional): Search keyword in title or venue.
 - `status` (string, optional): `upcoming`, `ongoing`, `completed`.

Success Response (`200 OK`)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Events retrieved successfully",
  "data": {
    "items": [
      {
        "id": "e2a1496a-04b3-4f98-904d-17bb27a7c501",
        "title": "BSIT Tech Summit 2026",
        "slug": "bsit-tech-summit-2026",
        "venueName": "Gymnasium, Main Campus",
        "startDate": "2026-11-15T08:00:00.000Z",
        "endDate": "2026-11-15T17:00:00.000Z",
        "totalRegistrations": 342,
        "totalCheckins": 289
      }
    ],
    "pagination": {
      "currentPage": 1,

```

```
    "totalPages": 1,  
    "totalItems": 1,  
    "limit": 10  
  }  
}  
}
```

3.3 Get Event Details

Retrieves full details for a single event by ID or slug.

- **URL:** `/events/:id`
- **Method:** `GET`
- **Access Control:** Public

Success Response (`200 OK`)

```
{  
  "success": true,  
  "statusCode": 200,  
  "message": "Event details fetched successfully",  
  "data": {  
    "id": "e2a1496a-04b3-4f98-904d-17bb27a7c501",  
    "organizerId": "a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11",  
    "title": "BSIT Tech Summit 2026",  
    "slug": "bsit-tech-summit-2026",  
    "description": "Annual student technology conference and  
project showcase.",  
    "venueName": "Gymnasium, Main Campus",  
    "startDate": "2026-11-15T08:00:00.000Z",  
    "endDate": "2026-11-15T17:00:00.000Z",  
    "isActive": true,  
    "checkpointsCount": 5,  
    "registrationsCount": 342  
  }  
}
```

```
}  
}
```

3.4 Update Event

Modifies event attributes, schedule, or venue.

- **URL:** `/events/:id`
- **Method:** `PUT`
- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER`)

Request Body

```
{  
  "title": "BSIT Tech Summit 2026 (Updated)",  
  "venueName": "University Theater",  
  "endDate": "2026-11-15T18:00:00.000Z"  
}
```

Success Response (`200 OK`)

```
{  
  "success": true,  
  "statusCode": 200,  
  "message": "Event updated successfully",  
  "data": {  
    "id": "e2a1496a-04b3-4f98-904d-17bb27a7c501",  
    "title": "BSIT Tech Summit 2026 (Updated)",  
    "venueName": "University Theater",  
    "updatedAt": "2026-10-06T14:00:00.000Z"  
  }  
}
```

3.5 Delete Event

Archives or soft-deletes an event.

- **URL:** `/events/:id`
- **Method:** `DELETE`
- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER`)

Success Response (`200 OK`)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Event successfully deleted"
}
```

3.6 Assign Staff to Event / Checkpoint

Assigns an organizer or scanner staff member to an event and binds them to a specific scanner station.

- **URL:** `/events/:id/staff`
- **Method:** `POST`
- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER`)

Request Body

```
{
  "userId": "c5f1190e-436d-4952-b8ec-f2323fcf912b",
  "assignedCheckpointId": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d"
}
```

Success Response (`201 Created`)

```
{
  "success": true,
  "statusCode": 201,
  "message": "Staff member assigned successfully",
  "data": {
    "id": "78129031-fb21-46bb-8bc2-a1089201bd29",
    "eventId": "e2a1496a-04b3-4f98-904d-17bb27a7c501",
    "userId": "c5f1190e-436d-4952-b8ec-f2323fcf912b",
    "assignedCheckpointId": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
    "createdAt": "2026-10-06T14:05:00.000Z"
  }
}
```

4. Checkpoint Management (/checkpoints)

4.1 Create Checkpoint

Defines a verification station for an event (Main Entrance, Workshop Session, Project Booth, or Exit Gate).

- **URL:** /events/:eventId/checkpoints
- **Method:** POST
- **Access Control:** Authenticated (ADMIN , ORGANIZER)

Request Body

```
{
  "name": "Main Entrance Gate A",
  "type": "MAIN_ENTRY",
  "description": "Primary gate verification for registered attendees.",
  "allowMultipleScans": false,
}
```

```
"isActive": true
}
```

Checkpoint Types

- **MAIN_ENTRY** : Primary event entry gate (defaults to single scan).
- **SESSION** : Specific talk or workshop session inside the venue.
- **BOOTH** : Project exhibition booth or sponsor booth (multi-scan).
- **EXIT** : Gate exit check-out log.

Success Response (201 Created)

```
{
  "success": true,
  "statusCode": 201,
  "message": "Checkpoint created successfully",
  "data": {
    "id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
    "eventId": "e2a1496a-04b3-4f98-904d-17bb27a7c501",
    "name": "Main Entrance Gate A",
    "type": "MAIN_ENTRY",
    "allowMultipleScans": false,
    "isActive": true,
    "createdAt": "2026-10-06T14:10:00.000Z"
  }
}
```

4.2 List Event Checkpoints

Retrieves all verification checkpoints configured for an event.

- **URL:** `/events/:eventId/checkpoints`
- **Method:** `GET`

- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER` , `STAFF`)

Success Response (`200 OK`)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Checkpoints retrieved successfully",
  "data": [
    {
      "id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
      "name": "Main Entrance Gate A",
      "type": "MAIN_ENTRY",
      "allowMultipleScans": false,
      "totalScans": 289
    },
    {
      "id": "3c84f67e-128a-4cd9-bf71-9231fcf982aa",
      "name": "Web Development Workshop",
      "type": "SESSION",
      "allowMultipleScans": false,
      "totalScans": 84
    },
    {
      "id": "771e892c-091a-4ab8-990a-112233445566",
      "name": "AI Innovation Booth",
      "type": "BOOTH",
      "allowMultipleScans": true,
      "totalScans": 156
    }
  ]
}
```

4.3 Update Checkpoint

Modifies checkpoint configuration or multi-scan permissions.

- **URL:** `/checkpoints/:id`
- **Method:** `PUT`
- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER`)

Request Body

```
{
  "name": "Main Entrance Gate A & B",
  "allowMultipleScans": false
}
```

Success Response (`200 OK`)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Checkpoint updated successfully",
  "data": {
    "id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
    "name": "Main Entrance Gate A & B",
    "updatedAt": "2026-10-06T14:15:00.000Z"
  }
}
```

5. Attendee Registration & RSVP (`/registrations`)

5.1 Public Event Registration / RSVP

Allows attendees to register for an event, automatically issuing a unique ticket code and QR code payload.

- **URL:** `/events/:eventId/register`

- **Method:** POST
- **Access Control:** Public

Request Body

```
{
  "firstName": "Alen Mikel",
  "lastName": "Umandal",
  "email": "amc.umandal@gmail.com",
  "studentOrIdNumber": "2023-01982-MN-0",
  "affiliation": "Department of Information Technology",
  "ticketType": "STUDENT",
  "customData": {
    "dietaryRequirements": "None",
    "emergencyContact": "09171234567"
  }
}
```

Success Response (201 Created)

```
{
  "success": true,
  "statusCode": 201,
  "message": "Registration confirmed. QR ticket generated.",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "eventId": "e2a1496a-04b3-4f98-904d-17bb27a7c501",
    "ticketCode": "CHK-2026-9812A",
    "firstName": "Alen Mikel",
    "lastName": "Umandal",
    "email": "amc.umandal@gmail.com",
    "studentOrIdNumber": "2023-01982-MN-0",
    "ticketType": "STUDENT",
    "status": "CONFIRMED",
  }
}
```

```
    "qrPayload": "CHK:e2a1496a:CHK-2026-9812A",
    "qrCodeUrl": "<https://api.checkpoint.site/api/v1/registrations/550e8400-e29b-41d4-a716-446655440000/qr>",
    "createdAt": "2026-10-06T14:20:00.000Z"
  }
}
```

Error Response (409 Conflict)

```
{
  "success": false,
  "statusCode": 409,
  "error": "DUPLICATE_REGISTRATION",
  "message": "An active registration with this email already exists for this event."
}
```

5.2 Lookup Ticket by Code

Fetches attendee registration and check-in status by unique ticket code.

- **URL:** `/registrations/:ticketCode`
- **Method:** `GET`
- **Access Control:** Public / Scanner App

Success Response (200 OK)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Ticket found",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "ticketCode": "CHK-2026-9812A",
  }
}
```

```

    "firstName": "Alen Mikel",
    "lastName": "Umandal",
    "email": "amc.umandal@gmail.com",
    "studentOrIdNumber": "2023-01982-MN-0",
    "affiliation": "Department of Information Technology",
    "ticketType": "STUDENT",
    "status": "CONFIRMED",
    "firstCheckedInAt": "2026-11-15T08:14:22.000Z",
    "event": {
      "title": "BSIT Tech Summit 2026",
      "venueName": "Gymnasium"
    }
  }
}

```

5.3 Get Attendee Attendance History

Retrieves all checkpoint stops, workshops, and booths visited by the attendee.

- **URL:** `/registrations/:ticketCode/history`
- **Method:** `GET`
- **Access Control:** Public / Authenticated

Success Response (`200 OK`)

```

{
  "success": true,
  "statusCode": 200,
  "message": "Attendance history fetched",
  "data": {
    "ticketCode": "CHK-2026-9812A",
    "attendeeName": "Alen Mikel Umandal",
    "scans": [
      {
        "checkpointName": "Main Entrance Gate A",

```

```

    "type": "MAIN_ENTRY",
    "scannedAt": "2026-11-15T08:14:22.000Z",
    "status": "SUCCESS"
  },
  {
    "checkpointName": "Web Development Workshop",
    "type": "SESSION",
    "scannedAt": "2026-11-15T10:05:10.000Z",
    "status": "SUCCESS"
  },
  {
    "checkpointName": "AI Innovation Booth",
    "type": "BOOTH",
    "scannedAt": "2026-11-15T13:45:00.000Z",
    "status": "SUCCESS"
  }
]
}

```

6. QR Code Verification & Scanning (`/checkpoints/:id/scan`)

6.1 Process QR Code Scan

Core check-in endpoint called by staff scanners upon capturing an attendee's QR code.

- **URL:** `/checkpoints/:checkpointId/scan`
- **Method:** `POST`
- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER` , `STAFF`)

Request Body

```
{
  "ticketCode": "CHK-2026-9812A",
  "deviceInfo": "Scanner Station #1 - iPad Air Safari",
  "notes": ""
}
```

Success Response (**200 OK** - Valid Check-In)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Check-in successful",
  "data": {
    "scanResult": "SUCCESS",
    "scannedAt": "2026-11-15T08:14:22.000Z",
    "checkpoint": {
      "id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
      "name": "Main Entrance Gate A",
      "type": "MAIN_ENTRY"
    },
    "attendee": {
      "fullName": "Alen Mikel Umandal",
      "studentId": "2023-01982-MN-0",
      "ticketType": "STUDENT",
      "status": "CONFIRMED"
    }
  }
}
```

Error Response (**409 Conflict** - Duplicate Scan on Single-Scan Gate)

```
{
  "success": false,
```

```
"statusCode": 409,
"error": "DUPLICATE_SCAN",
"message": "Attendee has already checked in at this checkpoint.",
"data": {
  "scanResult": "DUPLICATE",
  "firstScannedAt": "2026-11-15T08:14:22.000Z",
  "attendee": {
    "fullName": "Alen Mikel Umandal"
  }
}
```

Error Response (404 Not Found - Invalid Ticket Code)

```
{
  "success": false,
  "statusCode": 404,
  "error": "INVALID_TICKET",
  "message": "Ticket code does not exist or is not valid for this event."
}
```

6.2 Get Checkpoint Scan Stream

Returns recent check-in logs recorded at a specific scanner station.

- **URL:** `/checkpoints/:checkpointId/scans`
- **Method:** `GET`
- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER` , `STAFF`)
- **Query Parameters:**
 - `limit` (integer, default: `20`): Number of recent scans to fetch.

Success Response (200 OK)

```
{
  "success": true,
  "statusCode": 200,
  "message": "Recent scans retrieved",
  "data": [
    {
      "id": "11982301-44bb-4211-bcde-990182736451",
      "ticketCode": "CHK-2026-9812A",
      "attendeeName": "Alen Mikel Umandal",
      "status": "SUCCESS",
      "scannedAt": "2026-11-15T08:14:22.000Z"
    }
  ]
}
```

7. Live Dashboard, Analytics & Reports (/events/:id/analytics , /reports)

7.1 Real-Time Event Attendance Overview

Fetches live event metrics including total registrations, check-in totals, turnouts, and hourly check-in rates.

- **URL:** /events/:id/analytics/live
- **Method:** GET
- **Access Control:** Authenticated (ADMIN , ORGANIZER)

Success Response (200 OK)

```
{
  "success": true,
  "statusCode": 200,
```

```

"message": "Live metrics retrieved",
"data": {
  "totalRegistered": 500,
  "totalCheckedIn": 412,
  "turnoutPercentage": 82.4,
  "currentInsideVenue": 380,
  "checkinVelocityPerHour": 124,
  "ticketTypeBreakdown": {
    "STUDENT": 350,
    "VIP": 25,
    "SPEAKER": 10,
    "REGULAR": 27
  }
}
}

```

7.2 Checkpoint Traffic Analytics

Returns attendance and scan counts segmented across individual checkpoints, workshops, and booths.

- **URL:** `/events/:id/analytics/checkpoints`
- **Method:** `GET`
- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER`)

Success Response (`200 OK`)

```

{
  "success": true,
  "statusCode": 200,
  "message": "Checkpoint analytics retrieved",
  "data": [
    {
      "checkpointId": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
      "name": "Main Entrance Gate A",

```

```

    "type": "MAIN_ENTRY",
    "uniqueAttendees": 412,
    "totalScans": 412
  },
  {
    "checkpointId": "3c84f67e-128a-4cd9-bf71-9231fcf982aa",
    "name": "Web Development Workshop",
    "type": "SESSION",
    "uniqueAttendees": 120,
    "totalScans": 120
  },
  {
    "checkpointId": "771e892c-091a-4ab8-990a-112233445566",
    "name": "AI Innovation Booth",
    "type": "BOOTH",
    "uniqueAttendees": 215,
    "totalScans": 340
  }
]
}

```

7.3 Export Attendance Log

Generates and downloads a complete attendance report in CSV or PDF format.

- **URL:** `/events/:id/reports/export`
- **Method:** `GET`
- **Access Control:** Authenticated (`ADMIN` , `ORGANIZER`)
- **Query Parameters:**
 - `format` (string, required): `csv` , `pdf` , or `xlsx` .
 - `checkpointId` (UUID, optional): Filter export to a specific checkpoint.

Headers Returned

Content-Type: text/csv

Content-Disposition: attachment; filename="bsit-tech-summit-2026-attendance.csv"